



Session : Flight Planning

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Flight Planning

Flight Planning

Introduction

The OFP is prepared by Flight Dispatch for all Gulf Air flights. The OFP is a regulatory document and shall satisfy the requirement of BCAA and GF Operations Manual.

For full explanation of GF Operational Flight Plans refer to *OMA 8.1.10.9.1*.

The Flight Dispatcher must analyse all the factors and check all data that is required for the preparation of the flight plan before releasing the OFP for safe, legal and economical operations.

Flight Planning

Introduction

Contents included in the procedures:

1. Flight List Page
2. OFP Preparation Time
3. Checking Flight Plans Data
4. Operating Minima for Origin, Destination & Alternate Airports
5. Generating Flight Plans
6. Checking Flight Plans Output
7. Distribution of Briefing Package
8. ZFW Change after OFP Produced
9. Destination & Alternats Minima Guidance Table
10. Checklist for OFP Preparation

Flight Planning

Introduction

Flight List Page

To access the list page, for all flights for the date, the Flight Planning Client needs to be opened. Each Flight Dispatcher should use his/her own allocated User ID and Password for accessing the FPM.

By opening the flight plan client, the List Page for all flights will be displayed. Data read from other systems (MM, AIMS, FALCON & CGO System) that will be used to process flight plan calculation are available through interface. Missing data shall be checked by the Dispatcher from the source of the information to be updated in the flight planning module of FPM.

Flight Planning

Introduction

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Flight Planning

Introduction

Load

The Passenger (Pax) and Freight (Frgt) figures are taken from Falcon and Cargo system automatically at specified time intervals. When the load information taken through the interface is different from the load reported by the respective station, insert the station EZFW figure in the appropriate field.

Changing Dry Operation Weight

to reduce DOW whenever there is a ferry flight or a flight with reduced catering, the following pantry codes are added:

- A – Reduces the DOW by 500 kg (All aircraft types)
- B – Reduce the DOW by 1000 kg (All aircraft types)
- C – Reduces the DOW by 1500 kg (All aircraft types)
- D – Reduces the DOW by 2000 kg B787
- W - Increased DOW portable water to 100%
- A320 by - 50 kg
- A787 by - 175 kg

The default “@” is standard DOW as issued by MOT.

Flight Planning

Introduction

Aircraft MEL/CDL

If an aircraft is under MEL/CDL Dispatcher shall take appropriate action in the MEL List page of the FPM system. Dispatcher MEL entry is valid only for one time use. To Edit or Enter MEL data:

- Select "MEL" page'
- To ignore MEL implication select "Ignore" button for the appropriate MEL list
- To add new MEL, click "ADD" button to open the MEL dialog page, then
- Select the appropriate MEL from the list and click "OK" to activate

Flight Dispatcher shall cross check the MEL/CDL penalty against the aircraft manual at all time and any discrepancy shall be reported to Navigation Services for corrective action. Also cross check MEL/CDL with status sheet and log book. Contact MCC for any clarification. Items affecting the Load Sheet calculation may addressed to Load Sheet personal.

Flight Planning

Operating Minima for Origin, Destination & Alternate Airports

Reference: OMA 8.1.2.3.2

OMA 8.1.2.3.3

OMA 8.1.2.3.4

OMA 8.1.3.3

Ipad Jeppesen Airway application.

Flight Planning

Operating Minima for Origin, Destination & Alternate Airports

Take-Off Alternate

- When performance or weather condition prevents a flight returning to the departure airport, a Take-off alternate must be nominated on the OFP. The landing minimum for departure airport is the applicable minima for approach and landing.
- Weather at Take-off alternate must be at or above the applicable landing minimums, during the period commencing 1 hour before and ending 1 hour after the expected ETA at the airport.
- Maximum distance for Take-off alternate shall not exceed specified distance for the aeroplane type, ref OMA 8.1.2.3.2

Flight Planning

Operating Minima for Origin, Destination & Alternate Airports

Take-Off Alternate

-Twin-engine aeroplanes are restricted to 1 hr. flying time at single engine cruising speed. For EDTO certified operation, this can be extended to the maximum authorized EDTO diversion time with condition of crew qualification and EDTO maintenance check for the aircraft.

-Normally one or more of the alternates nominated in the 'Destination/Alternates' listings will be within the required range for use as a take-off alternate. If not, a take-off alternate should be nominated. For EDTO approved and qualified flight, the take-off alternates can be extended up to the maximum approved EDTO threshold distance.

-In the absence of suitable Take-off alternate airport, flight will be forced to be delayed until the origin airport weather improves. Refer 8.1.2.3.2 Take-off Alternate Aerodrome.

Flight Planning

Operating Minima for Origin, Destination & Alternate Airports

Destination Minima

Destination airport is considered suitable, if the weather reports or forecast for the destination indicate that from 1 hr. before to 1 hr. after ETA, it is above the applicable minima as per OMA 8.1.2.3.4 item 1.

Ignore suitability been activated, such option will be used in events were the weather report is not restricting Gulf Air operation. But, FPM is ignoring such station and considers it as below minima.

Flight Planning

Operating Minima for Origin, Destination & Alternate Airports

Destination Minima

-The applicable minima are:

-For precision approach, RVR/visibility specified in accordance with the applicable airport operating Minima (AOM). (Refer Concept of Minima in OMA - 8.1.3.3)

-For a non-precision approach, VIS/RVR and Ceiling at or above MDH.

-If the destination weather does not meet the above minima or nil weather for destination, two suitable alternates must be selected, and the most distant one to be nominated as primary alternate on the OFP for fuel calculation.

Flight Planning

Operating Minima for Origin, Destination & Alternate Airports

Destination and En-route Alternates Minima

A destination alternate shall be specified for all flights prior to take-off unless it satisfies the conditions prescribed in OMA 8.1.2.3.4. The requirements for selection of at least one or two alternates is explained as follows:

- At least one destination alternate to be specified on the OFP request provided destination weather is above the applicable minima.
- Two usable destination alternates to be nominated if the weather reports or forecast for the destination indicate that from 1 hour before to 1hour after ETA, it is below the applicable minima or no weather information for destination.
- For escape route flights, en-route alternate ports should be above en-route minima.

Flight Planning

Operating Minima for Origin, Destination & Alternate Airports

Destination and En-route Alternates Minima

Planning Minima for Destination & En-route Alternate

TYPE OF APPROACH	PLANNING MINIMA/RVR
CAT II & CAT III	CAT I (Note 1)
CAT I	Non-Precision (Notes 1 and 2)
Non-precision	Non-Precision (Notes 1 and 2) plus 200 ft/1,000m

Note 1: RVR

Note 2: The ceiling must be at or above MDH.

'Non precision minima' in the above Table, means the less restricted minimum that is available in the prevailing wind and serviceability conditions; Localizer Only approaches, if published, are considered to be 'non precision' in this context.

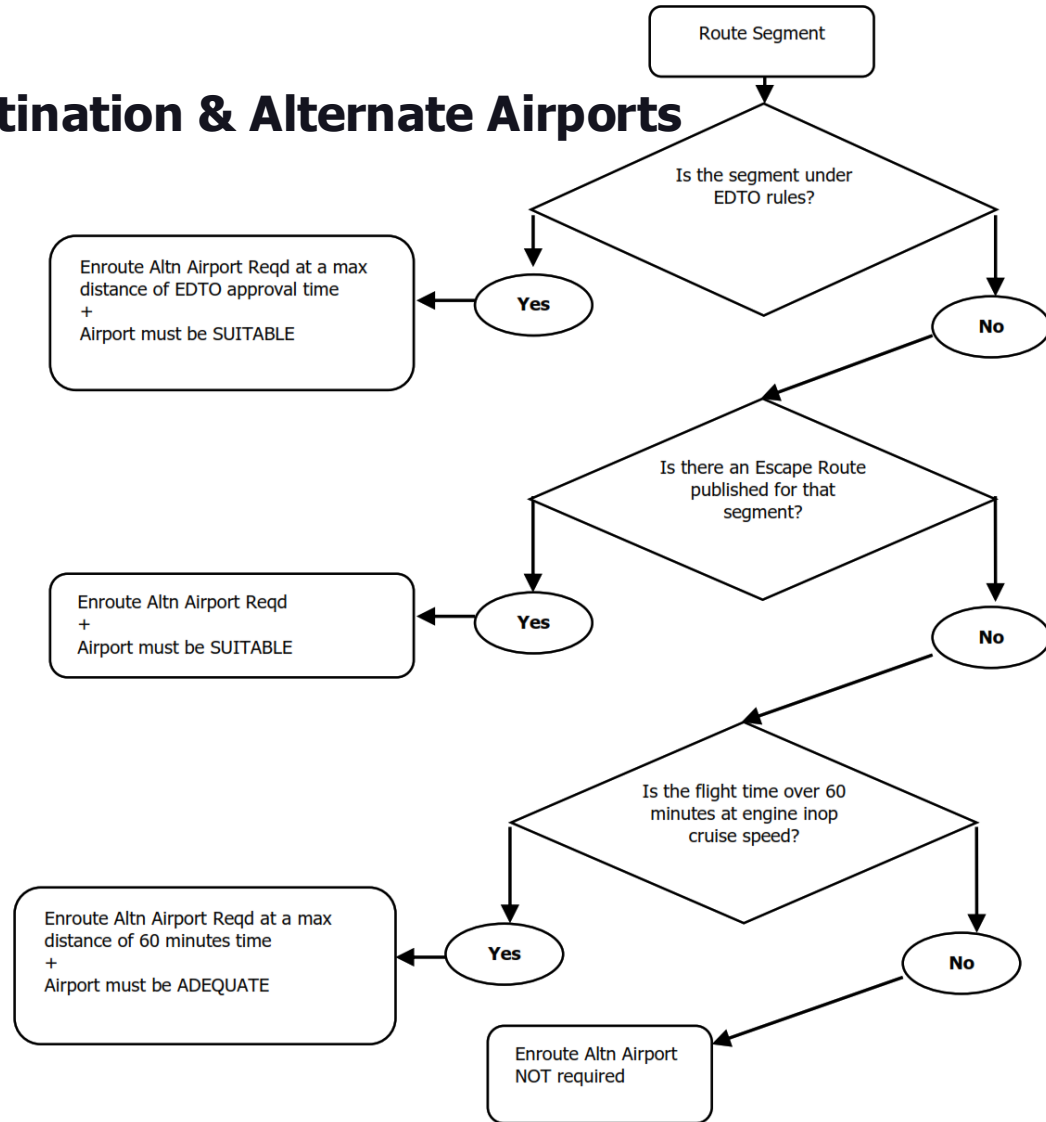
When two alternates are required because the weather forecast for the destination aerodrome is below landing minima or not available, fuel to reach the most distant alternate must be carried. Ref OMA 8.1.2.3.5.3.

Flight Planning

Operating Minima for Origin, Destination & Alternate Airports

Need for an Enroute Alternate

Ref. OMA 8.1.2.3.3; Not all flights need Enroute alternates to be specified. Each segment has to be analyzed base on the following flow chart



Flight Planning

Generating Flight Plans

Available Route

If routes are stored for selected leg, the Available Routing dropdown list will display the list of routes available for the chosen city-pair.

The system is set for auto calculation (STD-10 Hrs) by going to the Variant page – load all available routes and calculate.

Do the following to select the most convenient route for the day:

- a) Check availability of data:
 - MM information (Reg, Leg, STD)
 - Load
 - MEL/CDL if reported
 - Tankering
 - NOTAM – CF (Company Fuel)
 - WX (Airports and En-route)
 - Alternate

Flight Planning

Generating Flight Plans

Available Route

- b) Go to "VARIANT" page
- c) Select "Cr All" then "OK" (to delete existing variants)
- d) Select "CP Comp" (City Pair Comparison)
- e) Select the best route. (the variant with '*' is the active route) avoid 99D route
- f) Go to "Flight" page
- g) Tick the "EROPS" box & recalculate

Flight Planning

Generating Flight Plans

Creating and Editing Route

This page will allow Dispatchers to create and/or edit and optimize routings for given city-pair and en-route flight planning when it is required. Selection of waypoints to join departure and arrival ports; option to avoid routing; waypoints and FIR can be processed on this page.

NOTAMs

The NOTAM page provides NOTAM Briefing for selected flight or chain of flights. Additional NOTAMs can be added by the Dispatcher by selecting the four letter code for the airport and/or FIR if required. To view the NOTAM, the flight must always be calculated to get the routing for collection of valid NOTAMs for the route.

-NOTAMS for First and Second alternate airport should always be added to the 'BP' even if they're not nominated as alternate airports.

-If the airport is not DST,ORG,ALTs or ERA ; then the airfield NOTAM will not be automatically reflected in the briefing package.

Flight Planning

Generating Flight Plans

Weather:

By selecting the Met Page, you can access the actual METAR and Forecast Weather information for departure, en-route, destination and destination alternates and any other airports required to be included. The Dispatcher shall check the METAR/TAF of the respective airports to monitor the automatic selection of suitable alternates by the flight planning system.

The en-route WX information display by the system is only for those airports that gives 60 minutes EROPS coverage, and any additional WX information must be added manually when required.

Flight Planning

Generating Flight Plans

Weather

The SIG WX Chart shall be checked and analysed to make sure that each flight is routed via a safe en-route weather phenomena.

Applicable SIG-Chart valid for the route shall be generated to analyse the en-route weather.

The following are area of operations and Name/Description of SIG Chart that are recommended to be used to analyse the WX:

Name	Description
PGRE05	Africa - WAFC London
PGKE05	Antarctic - WAFC London
PGJE05	Antarctic - WAFC Washington
PGCE05	Asia & Europe - WAFC London
PGZE05	Asia - WAFC London
PGIE05	Atlantic Basin - WAFC Washington
PGDE14	Europe - WAFC London
PGSE05	Europe to South America - WAFC London
PGZE14	India - WAFC London
PGCE14	Middle East - WAFC London
PGNE14	NAT - WAFC Washington
PGAE05	North Atlantic, Europe & USA - WAFC London
PGAE05	North Atlantic, Europe & USA - WAFC Washington
PGBE05	North Pacific to East Asia - WAFC Washington
PGDE29	Northern Pacific - WAFC Washington
PGEE05	The Americas - WAFC Washington
PGGE05	Tropical Belt - WAFC London
PGGE05	Western Pacific - WAFC Washington

-Avoid significant WX activity area like FREQCB, Tropical Storm & Tropical Cyclone.

Flight Planning

Generating Flight Plans

Alternate Airport Selection

Take-off, En-route and Destination alternates that is selected by the system automatically shall be cross checked. When there is a need for the Dispatcher to alter these alternates to fulfil different requirements, the same can be selected manually by accessing the respective field.

Tankering/Non-Tankering

Flight Planning

Fuel Tankering

Fuel tankering is indicated by either a plus (+) or minus (-) value of dollar for carrying 1000 kg of fuel between departure and destination airports. The **FEco** is the indicator for Fuel tankering that will be displayed with a "+/-" cost sign for approximate saving. The 'Target' field should read 'Auto' and the system is set to calculate the tankering legs STD-10 hrs.

Whenever there is a fuel saving (+) between the first and consecutive stations, the system will automatically analyse the fuel required for the next sectors and display the remaining fuel at the next airport in the 'remF' field of the Flight Page.

Flight Planning

Fuel Tankering:

Fuel Tankering Procedures

Flight Dispatchers should always consider the uplift of extra fuel for the next sectors (without incurring payload penalty) whenever feasible. FPM automatically calculates whether tankering will be cost-effective or not.

When multi-sector tankering is calculated, payload figures taken from the system at the time of planning may not be accurate. This inaccuracy of payload on the flight plan may restrict the subsequent airports loads, if the OFP shows Maximum Limited Take-off Weight (TOW) or Landing Weight (LGW). Dispatchers are to be vigilant on such sectors and reduce the Tankering fuel to leave room for extra payload at the next airports. This applies particularly to multi sector Intra Gulf, to Indian sub-continent and to Cairo flights.

Refer FLIGHT DISPATCH PROCEDURE: FDP-14.

Flight Planning

MLW Buffer for Tankering Sector

For tankring sector you are requested not to tanker up to MLW, below buffers shall be kept in consideration.

A320/321 REDUCE MLDW BY 600KGS

B787 REDUCE MLDW BY 1000KGS.

Flight Planning

Nonstandard Changes

Whenever there is a requirement to add extra fuel due to NOTAM or the fuel in the aircraft tank is more than the minimum required, the additional fuel should be included on the flight plan calculation.

Dispatchers are encouraged to utilize the space provided in the OFP to write any non-standard changes in the OFP and shall record it in the log. This will help answering future inquiries and avoid unnecessary emails to provide details.

Examples of nonstandard changes;

- Any changes resulted in extra fuel
- Using longer alternate
- CAPT request
- Choosing Non optimum Route usage
- RCFP

Flight Planning

Take –off and Landing Weight Performance (TOW/LDW)

FPM will use in future, the Standard Computerized Aircraft Performance program (SCAP) and forecast or predicted weather data to determine the Max TOW/LDW calculation for system selected runway based on the wind report.

However, in the case of operations, like SYZ and SAH, where LDW restriction is based on the worst TOW figure at the airport, a manual entry of LDW value is required before the flight plan calculation.

- Open the 'Flight Page'
- Go to 'Aircraft Weights and Limits' box
- Select 'LDW' box on the right column (left column is actual value)
- Enter the 'LDW' figure for the station and aircraft type as required to limit maximum landing weight.
- Calculate flight plan

Flight Level Restrictions

Flight Planning

MINIMUM FLIGHT ALTITUDES

VFR Flights

All commercial flights shall be conducted under IFR rules. Only non-revenue, demonstration, test and special flights under VFR may be permitted by the DFO.

For flights conducted under VFR rules, except when necessary for take-off or landing, an aircraft may not operate below the following altitudes:

1. Over congested areas – over any congested area of a city or town, an altitude of 1,000 ft. above the highest obstacle within a horizontal radius of 2,000 ft. of the aircraft.
2. Over areas other than congested areas – An altitude of 500 ft. above the surface, except over open water or sparsely populated areas. In those cases, the aircraft may not be operated closer than 500 ft. to any person, vessel, vehicle, or structure.

Flight Planning

MINIMUM FLIGHT ALTITUDES

VFR Flights

3. Over an open air assembly of persons –at a height as will permit, in the event of an emergency arising, a landing to be made without undue hazard to persons or property on the surface.

4. Over an open air assembly of persons – at an altitude from which, in the event of an emergency necessitating an immediate landing, it would be possible to land the aircraft without creating a hazard to persons or property on the surface.

Flight Planning

MINIMUM FLIGHT ALTITUDES

IFR Flights

ANTR OPS 1.365

In accordance with the basic requirements of ANTR-OPS 1.365, except for take-off/ departure or approach/landing, all Gulf Air airplanes shall be operated at altitudes which clear all obstacles by at least:

1. 2,000 ft. vertically over high terrain or in mountainous areas, within a radius of 4 NM from the estimated position of the airplane; and
2. 1,000 ft. vertically over terrain elsewhere than as described in the bullet above, within a radius of 4NM from the estimated position of the airplane.

High Terrain is defined as an obstacle or terrain exceeding 5,000 feet.

Flight Planning

MINIMUM FLIGHT ALTITUDES

IFR Flights

The estimated position of the airplane will take into account the navigation accuracy that can be achieved on the relevant route segment, considering the navigation equipment available on the airplane and the facilities available on the ground.

Additionally, all Gulf Air airplanes will be flown at or above altitudes required to comply with the performance requirements of paragraphs 8.1.1.6 and 8.1.1.7. (Enroute Performance Requirements)

Paragraph 8.1.1.3 describes how these minimum flight altitude requirements will be applied when using Jeppesen or Company produced navigation charts.

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MINIMUM FLIGHT ALTITUDES

IFR Flights

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Flight Planning

MINIMUM FLIGHT ALTITUDES

Presentation and Application of Minimum Flight Altitudes

During both normal and emergency operations, the following minimum altitudes shall be applied:

Terminal Areas

The minimum altitude information for terminal charts is given on the

- Expanded Area Chart
- Instrument Approach Procedure chart
- Radar Minimum Altitudes Chart
- SID or STAR chart

Flight Planning

MINIMUM FLIGHT ALTITUDES

Terminal Areas

If the airplane is within 25 nautical miles of an aerodrome, or such other range as may be published on the above charts, then it shall maintain an altitude not below the following minimum altitudes:

1. When established on a published SID, STAR, or Instrument Approach Procedure, the minimum flight altitude(s) as published for the procedure.
2. When being vectored under Radar Control, then the minimum altitude shall be the Minimum Vectoring Altitude (MVA) for that sector. It must be noted that the MVA may be only be utilized after the radar controller's advice that the aircraft is identified on radar by the term "Radar Contact" and with clearance to descent to that altitude. In case the descent to, or maintenance of the MVA altitude will infringe the sector boundaries, and the airplane may enter another sector with a higher MVA, then the clearance shall be challenged.
3. When the above bullets are not applicable, or when the airplane is at greater than 25 NM from the aerodrome or such other range as may be published on the above charts, then the enroute minimum flight altitude will apply.

Flight Planning

MINIMUM FLIGHT ALTITUDES

Enroute

The minimum altitude information for enroute segments is given on the Jeppesen charts. They contain the following minimum altitudes, whose descriptions are found in the Introduction pages of the Jeppesen Manual.

1. Minimum Enroute Altitude (MEA - altitude with no suffix)
2. Minimum Obstacle Clearance Altitude (MOCA - altitude suffix 'T')
3. Route Minimum Off Route Altitude (Route MORA - altitude suffix 'a')
4. Grid Minimum Off Route Altitude (Grid MORA)

These published minimum altitudes will be applied as follows:

1. When maintaining the centreline of an airway or ATS route, the minimum flight altitude will be the lowest of the:
 - a. Minimum Enroute Altitude, or
 - b. Minimum Off-route Altitude, or
 - c. Minimum Obstruction Clearance Altitude (only within 22 NM of the VOR).
2. Consideration shall be given to the ATC requirements and any foreseeable contingencies along the route

Flight Planning

MINIMUM FLIGHT ALTITUDES

Enroute

3. When deviating from the centreline of an airway or ATS route during normal flight operations, the applicable minimum flight altitude shall be the Grid MORA containing the aircraft's position. It must be noted that deviation from ATS routes is not permitted during normal flight operations where the Grid MORA is not published, unless a specific minimum flight altitude has been issued by DFO for that route and flight.
4. When deviating from an airway or ATS route for emergency reasons (i.e. decompression or drift down), and an 'Escape Route' is published (either the flight planning system, or by Navigation Services), and the aircraft is established on the specific escape route then the minimum altitude shall be the minimum authorized altitude for that segment of the Escape Route.
5. When deviating from an airway or ATS route for emergency reasons (i.e. decompression or drift down), and an 'Escape Route' is NOT published the minimum flight altitude shall be the Grid MORA.

Flight Planning

Flight below Minimum Altitudes

Safe Enroute Altitude/Minimum Safe Grid Altitude

Descent below the minimum safe enroute altitude/ minimum safe grid altitude to the Minimum Sector Altitude (MSA) / Terminal Area Altitude (TAA) may be made when approaching the navigation aid from which an approach-to-land will be conducted, provided the airplane's position can be accurately established as being within 25 NM from the navigation aid upon which the minimum sector altitude is based by:

1. The use of a radio navigation aids, or
2. Positive radar control.

Flight Planning

Flight below Minimum Altitudes

Descent below Minimum Sector Altitude / Terminal Area Altitude

The Minimum Sector Altitude (MSA) is the lowest altitude which will provide a minimum clearance of 1000 ft. above all objects located in an area contained within a sector of a circle of 25 NM radius centred on a radio aid to navigation.

The Terminal Area Altitude (TAA) replaces the MSA on some RNAV charts.

Descent below MSA/TAA is authorized only via a published instrument approach procedure or a visual approach, and only when so cleared by the appropriate ATS unit.

Flight Planning

Flight below Minimum Altitudes

Flight below Minimum Altitudes during Climb and Cruise

Airplanes proceeding in accordance with published departure routes are safe in respect of terrain and other restrictions on any level included in the departure route or otherwise assigned by ATC.

Minimum Safe Enroute Altitudes published for subsequent segments of a departure route shall be observed by commencement of climb sufficiently in advance to ensure that the point from which the higher minimum safe enroute altitude applies is crossed not below such altitude.

During an off-route departure, level flight may only be accepted at or above the MSA / TAA or minimum safe grid altitude. An airplane being radar vectored and positively identified may be flown below these altitudes if the pilot is able to monitor the airplanes position using the available radio navigational aids.

Airport Directory

Flight Planning

Airport Directory

OMC Manual

OMC contains specific instructions and information pertaining to navigation, communications and aerodromes within Gulf Air's approved area of operations.

The contents of Part C shall consist of material produced by Gulf Air, supplemented by the following publications:

1. Jeppesen Publications which include:

- Route Notes

- Region Specific Information

- i. Documents

- ii. Contingency

- iii. Reference

- Company Manuals

- i. Jeppesen Airway Manual – General

- ii. Jeppesen Airway Manual – Eastern Europe

- iii. Jeppesen Airway Manual – Pacific

- iv. Jeppesen Airway Manual – Europe

- v. Jeppesen Airway Manual – Africa

- vi. Jeppesen Airway Manual – Middle East

2. Enroute and Terminal Charts on Jepp FD PRO X application on the iPad / Terminal Charts.

3. Company briefing documentation produced by the Flight Dispatch Office.

Flight Planning

Airport Directory

FPM Airport Suitability Log

Airport Suitability Log will show a list of all flight-relevant airports checked by the system, for their different usages and time windows.

Airports, which are either used as, or available within the dropdown lists for departure, destination, departure/destination alternates, auxiliary airports, auxiliary alternates, as well as airports within the Quick EROPS field, will be checked for their suitability. The list is ordered alphabetically.

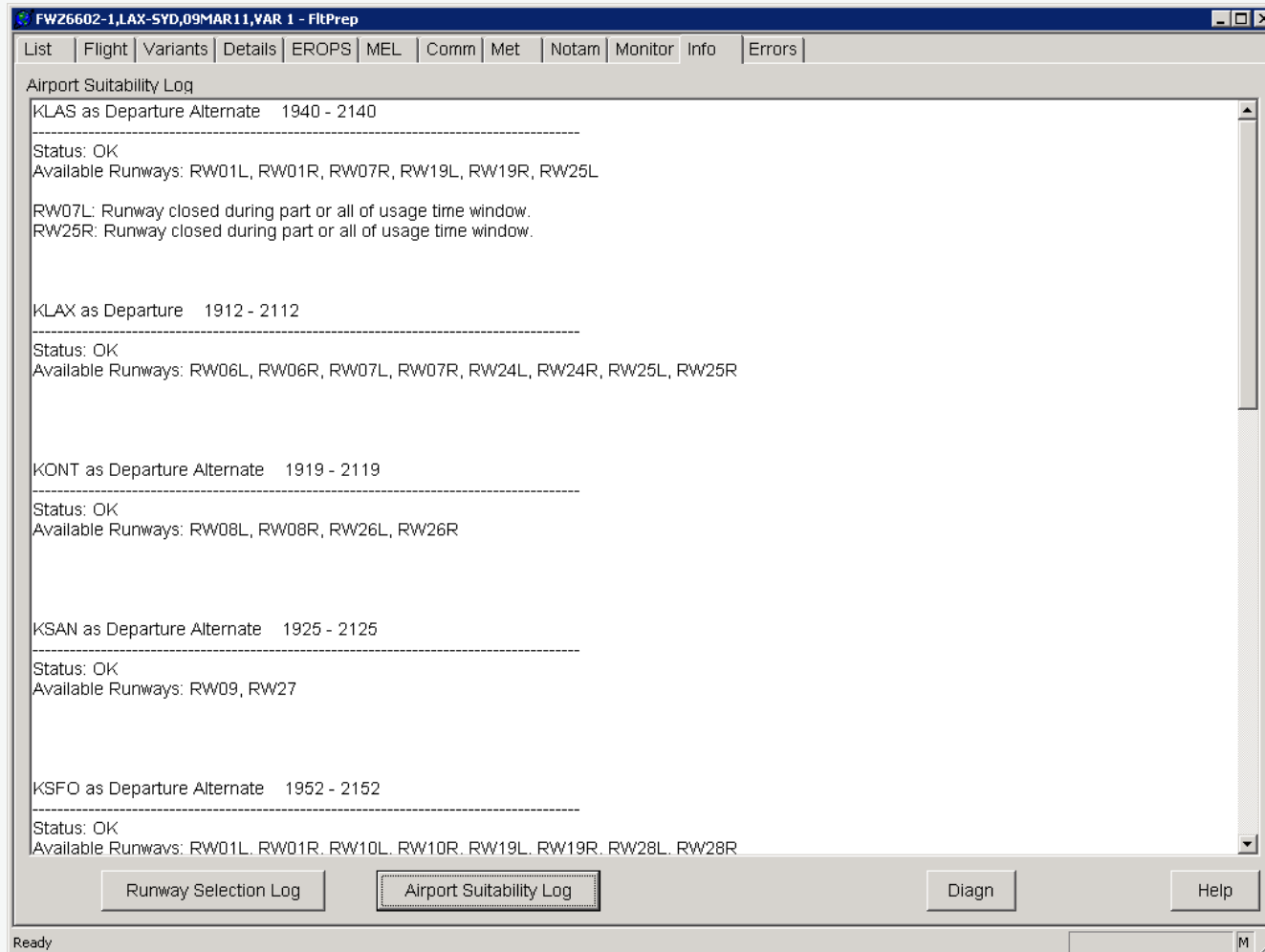
Checks include (depending on the configuration):

- airport usage authorization
- airport opening times
- runways longer than the assigned aircraft's minimum runway length
- weather forecast not below airport minima
- runways not exceeding crosswind and tailwind maxima
- NOTAM airport/runway closures

As conditions may change, the Airport Suitability Log will update each time the button is pressed.

Flight Planning

Airport Directory - FPM Airport Suitability Log



Thank you.

